



Integrating scientific and local knowledge to address conservation conflicts: Towards a practical framework based on lessons learned from a Scottish case study

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ABSTRACT

Integrating local knowledge with scientific knowledge can offer significant benefits to improving environmental decision-making. However, this is especially challenging in environmental conflict situations where advice is lacking, and no single approach can foster conflict transformation. To understand stakeholder knowledge and its integration in a conflict transformation process in Scotland, we brought together diverse stakeholder organisations and encouraged power sharing in the project's management. Our mixed-methods approach was based on theories of community science, knowledge co-production, knowledge integration and implementation and conflict transformation. We gathered stakeholder perceptions to see where local and scientific knowledge converged and diverged. Stakeholders holding opposing views mutually prioritised knowledge gaps and identified future collaborative actions. Building upon lessons learned, we present a practical framework and associated considerations to realise knowledge integration goals in conservation conflict situations. This framework is widely applicable, especially in situations where disputes over the evidence-base prevent positive outcomes for people and nature.

1. Introduction

Knowledge integration has long been advocated in policies such as the UN Convention on Biological Diversity, Agenda 21 of the Aarhus Conventions and the associated EU Directive on public participation, and platforms such as the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services (IPBES). Integrating local knowledge with scientific knowledge is increasingly thought to offer significant benefits to informing and improving environmental decision-making which needs to be based on salient, credible, and legitimate knowledge (O'Connell et al., 2013; Sarkki et al., 2014; Nguyen et al., 2017). The inclusion of local knowledge can provide the necessary socio-economic context for available science to inform specific decision contexts while validating decisions with scientific knowledge can increase the legitimacy of stakeholders' decision-making and foster

credibility, which is crucial for the acceptance of decisions among the wider community (O'Connor et al., 2019).

However, knowledge integration is challenging. The process requires accommodating different views and methodologies in a problem-solving, systematic, reflexive and cyclic manner (Raymond et al., 2010; Nguyen et al., 2017). Although principles for good science exist, similar methods for assessing the quality of local knowledge are lacking and practical challenges in accessing local knowledge (which is often tacit) make integration processes difficult to implement (Sillitoe, 2014; Sillitoe and Filer, 2014; O'Connor et al., 2019). Conversely, some stakeholders may mistrust scientific data sources, especially when social engagement and dialogue are lacking in the decision-making process (Cravens and Ardoin 2016). One criticism of knowledge integration is that since non-scientific knowledge is unique to its context, it faces loss of authenticity when merged with scientific knowledge because the

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process of generalising strips away essential local characteristics (Agrawal, 2002; Löfmarck and Lidskog, 2017), as has been found in some IPBES processes (Löfmarck and Lidskog, 2017).

Recognising and respecting a knowledge system has been likened to becoming fluent in a foreign language where words, cultural values and practices must all be learned and embraced (Löfmarck and Lidskog, 2017). This is problematic in conservation conflict situations where values and practices are central to the dispute and where stakeholders may interpret and portray research differently to support their own interests and agendas or undermine the position of others (Hodgson et al., 2018).

Conflicts are complex: underlying social, economic and ecological factors and uneven power dynamics drive division among interest groups, and entrenched views based on selective interpretation of scientific knowledge serve to maintain the status quo (Hodgson et al., 2019). Conflict transformation processes are necessarily lengthy and incremental because they entail questioning of values, beliefs and assumptions, and if systemic changes are large, they risk overwhelming those involved due to potential loss of identity (Berkes, 2017). Participatory and knowledge-based approaches to conflict transformation can be useful but are insufficient to fully transform conflicts (Stepanova, 2014). A new approach must be created through a process whereby researchers, managers, and stakeholders together define and redefine the questions, use a mix of study approaches, and deliberate on the relevant evidence (Ludwig, 2001; Berkes, 2017).

Taking the approach of knowledge integration further, one increasingly mainstream solution is the co-production of knowledge by researchers and practitioners, defined as an ‘iterative and collaborative process(es) involving diverse types of expertise, knowledge and actors to produce context-specific knowledge and pathways towards a sustainable future’ (Norström et al., 2020: 2).

The concept of ‘community science’ as place-based social learning¹ extends the principles of co-production. Here, knowledge sources are linked through a scientific mode of enquiry implemented by and within a given community who decides with whom to engage (Charles et al., 2020). Known benefits of community science approaches include developing comprehensive understanding and implementing joint next steps to address a problem; creating learning opportunities to cope with complexity and uncertainty; and building community resilience (Charles et al., 2020).

While effectively co-producing knowledge may be a policy goal and key to the success of conservation management programs, adequate resources and expertise are needed to achieve it. An implicit but unrecognised challenge is that conducting research involving different strands of knowledge requires skills among team members in research integration to develop a more comprehensive understanding of the problem and methods to address it, and research implementation to improve the situation (Bammer et al., 2020). Specifically, expertise is needed to recognise salient interconnections with other problems and knowing which methods to use in the given context (Bammer et al., 2020).

Despite the academic, political and social need, it appears that no framework for integrating knowledge currently exists in conservation conflicts, defined as situations that occur ‘when two or more parties with strongly held opinions clash over conservation objectives and when one party is perceived to assert its interests at the expense of another’ (Redpath et al., 2013: 100). Considering the impact of conservation conflicts on environmental conservation and human well-being (Redpath et al., 2013), a knowledge integration framework would be very useful in this context. That is, where acceptance of the validity of different types of knowledge bases is low, where stakeholders have opposing values and worldviews, and where levels of distrust can be high (Young et al., 2016).

We address this urgent gap by building on a novel, community-led knowledge co-production process that sought to integrate knowledge from scientists and practitioners embroiled in a conservation conflict in Scotland (United Kingdom). Through this process we develop a set of considerations based on the literature and lessons learned to address common challenges of knowledge integration processes, with key recommendations that can help achieve knowledge integration goals. We report on this process and outline a practical framework that can be implemented in situations where conflicts over the evidence-base currently prevent positive outcomes for people and nature.

2. Methods

2.1. Case study: community-led, knowledge co-production and integration in a conservation conflict

The current legislative framework for the protection of wildlife in Scotland is embedded in numerous EU Directives and domestic statutes which recognises that some species may impact on other interests. Heather moorlands provide high conservation value habitat for several ground-nesting bird species, though much is managed specifically for the game species red grouse (*Lagopus lagopus scoticus*) by controlling avian and mammalian predators (Thirgood and Redpath, 2008). The hen harrier (*Circus cyaneus*) is a protected predator species whose illegal killing on some grouse moors is the principal factor limiting population growth according to scientists (Murgatroyd et al., 2019). Many moorland stakeholders agree that ground-nesting bird populations are declining in Scotland but disagree about how best to manage this problem and disagree about the evidence behind different management options (Hodgson et al., 2018, 2019). The efficacy of habitat versus predator management for conserving bird populations is most contentious, leading to a high-profile conservation conflict between some conservationists and grouse moor managers (Young et al., 2016; Carrell, 2017).

In 2015, the Scottish Government funded the Moorland Forum (the Forum) to commission the Understanding Predation (UP) project to initiate a conflict transformation process. The Forum mandated a co-produced, knowledge integration approach by lead members of a research group (including the authors) who had delivered a scientific review for the Forum in 2005, but which was not widely used by policy and practice stakeholders (Park et al., 2005). They recognised that integration of ecological and local knowledge was essential to achieve a more holistic understanding of the issues, and to identify where joint action could be taken.

The project focused on six focal ground-nesting wild birds (the prey) and four of their common predators (Fig. 1). Forum members were particularly interested in these ten species: there were widely held concerns over recent population declines of the prey species and a shared desire to find ways of reversing declines, while the predator populations were thought to be increasing. Focusing on ground-nesting birds allowed us to take a step back from the more acute conflict around hen harrier conservation and grouse moor management, whilst still allowing for a discussion about conservation in moorlands. We aimed to build a publicly accessible evidence base collating and critically assessing available information from science and stakeholders’ local knowledge on predator-prey interactions, highlighting where these forms of knowledge agreed or disagreed, and analysing the reasons for any differences.

2.2. Data collection

A multi-stakeholder partnership comprising the Scottish Government, the Forum and scientists oversaw the project (Fig. 2). Initially, a planning group worked with Scottish Natural Heritage (the Scottish Government’s environment agency) to finalise the project plan, before dis-banding and forming the steering and research groups

¹ See Reed et al. (2010) for a definition of social learning.

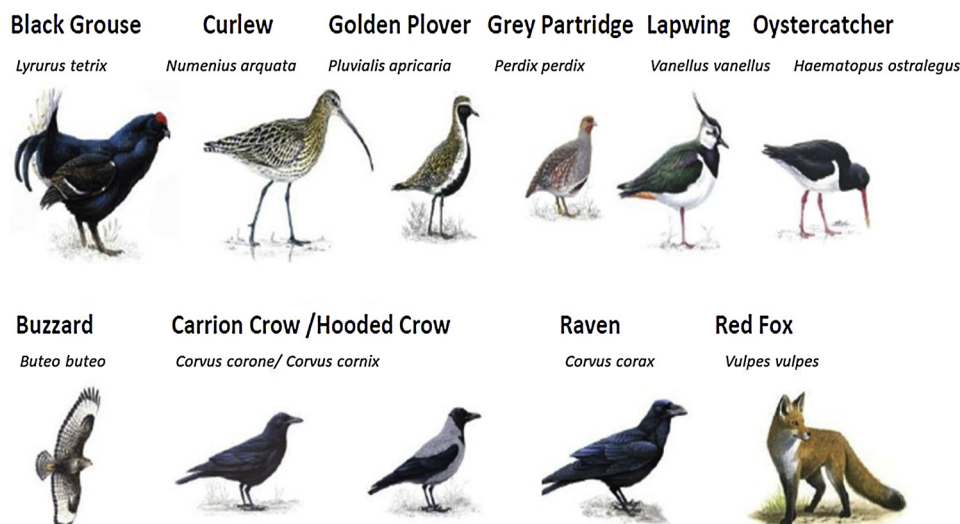


Fig. 1. Top: focal ground-nesting bird species consisting of four breeding wader and two game species (Black Grouse and Grey Partridge). Bottom: four common predators of the focal birds (Carrion and Hooded Crow were considered a single group ‘crows’) (adapted from Ainsworth et al., 2016).

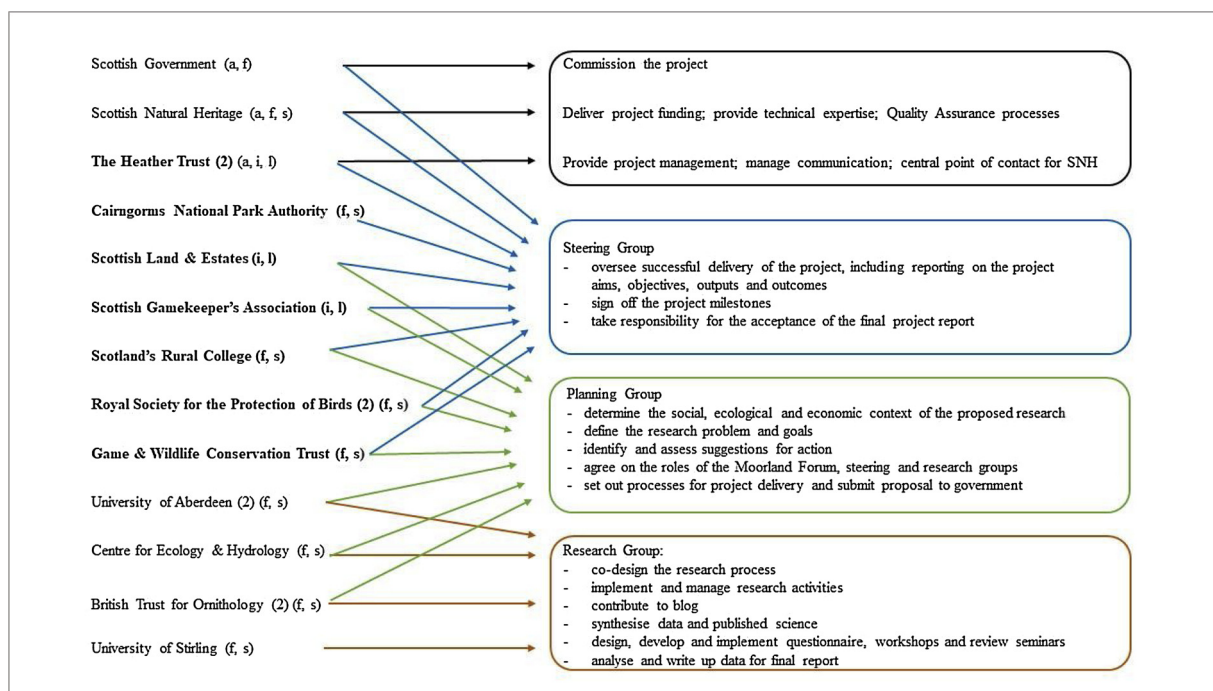


Fig. 2. Roles and responsibilities of partner organisations in the Understanding Predation project. Organisations are labelled based on their project role according to Stepanova's (2014) typology of knowledge for coastal conflict research, where a = administrative; f = formal; i = informal; s = scientific; and l = local knowledge. There was one representative from each organisation unless otherwise indicated in brackets.

(Fig. 2). We designed the process following overlapping principles for the practice of knowledge exchange to inform research, policy and practice (design, represent, engage, impact, reflect and sustain) (Reed et al., 2014) and co-production (context-based, pluralistic, goal-oriented, interactive) (Norström et al., 2020). This brought together diverse stakeholders who had previously interacted infrequently and among whom there were opposing views about managing the focal birds. We sought to map the conflict and explore stakeholder perceptions about management solutions to support conflict transformation (Redpath et al., 2013). This included identifying stakeholders; mapping their values, attitudes, goals and positions; gathering all scientific evidence together with gaps and uncertainties; identifying economic, ecological and social impacts; and understanding the wider socio-political context (e.g. legislation) (Redpath et al., 2013). We advanced this

process by integrating local knowledge with the scientific evidence to identify gaps and uncertainties.

2.2.1. Methodology

Our main objectives were to compare local and scientific knowledge regarding: i) the recent population changes of the ten focal prey and predator species; ii) evidence for the factors driving these changes (including predation); and iii) evidence for the success, or otherwise, of various types of conservation management for the prey species. The methods used to achieve these goals, and links between them are presented in Fig. 3. The main methods used are presented briefly below.

2.2.1.1. Literature reviews. The planning group identified a need to update and collate existing ecological knowledge to enable comparison

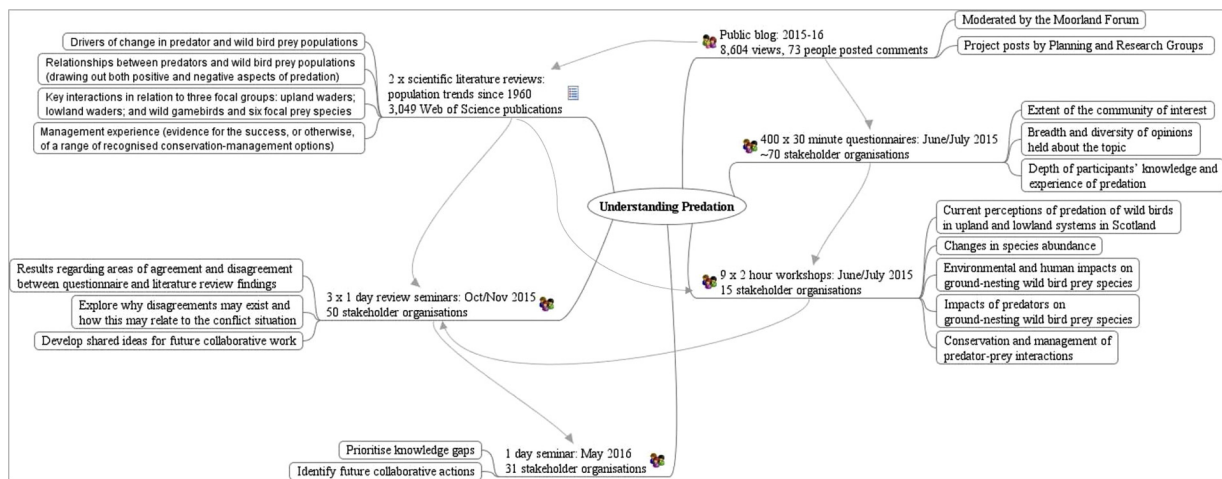


Fig. 3. Mind map showing the relationships between methods used in the Understanding Predation project. Bubbles indicate the objectives of each method and arrows indicate where one method contributed data directly to another.

of relevant aspects with stakeholders' local knowledge and to make this knowledge publicly accessible. Two natural scientists in the research group led two natural science literature reviews and the broader community was invited to contribute material via the blog and questionnaire. We synthesized data and published science on population trends since 1960, including a new Scottish and regional analysis of recent changes in wild bird populations. We also conducted a review of peer-reviewed publications gathered via Web of Science. The results are publicly available in [Ainsworth et al. \(2016\)](#).

2.2.1.2. Project blog. We needed to promote constructive discussion about the project because of the diverse and sometimes extreme views known to exist among community members about the management of predators. We therefore created a new public blog to explain the project's purpose, provide progress reports, and request community input regarding the questionnaire and focus group program and the literature review. Members of the public could voice their opinions about the project and its subject matter and learn the opinions of others in a non-confrontational way.

2.2.1.3. Questionnaire. Online surveys offer some advantages over other survey methods, including ease of creating and posting, low cost of administration, speed with which data can be gathered and low cost per respondent ([Gideon, 2012](#)). We therefore designed a structured, online questionnaire using blog comments and knowledge from planning, steering and research group members. The questionnaire linked to the natural science literature reviews, as it was our intention to discuss information from both the questionnaire and literature reviews in focus group discussions (see Supplementary Material [SM] for a copy of the questionnaire).

We piloted the questionnaire with members of the steering group and two independent scientists before distributing it in June/July 2015 to Forum members and relevant non-member organisations. We implemented a "snowball sampling" strategy via the blog, email and public events to address known problems associated with sampling concealed populations ([Atkinson and Flint, 2004](#). 1382 questionnaires were initiated with 400 being completed and analysed. Interested participants registered to join a focus group discussion.

2.2.1.4. Focus groups. For a more in-depth discussion with key stakeholders, we facilitated nine focus groups (June/July 2015) with 62 participants (two female) representing 15 organisations (see [O.Nyumba et al., 2018](#) for a useful overview of the methodology). This gender balance reflected the male-dominated nature of the activities carried out by stakeholders involved.

We invited stakeholders from like-minded organisations to attend each group (as the conflict was too acute to bring different organisations together at that stage), and organisational representatives nominated individual participants. We identified specific group interests from their collated questionnaire responses which participants deliberated upon together in the group setting. We also asked them to compare their questionnaire data with corresponding data from the literature reviews to understand their perceptions over potential disagreement between the two sources of knowledge. This novel approach created a more meaningful experience for participants while maintaining overall consistency regarding subjects covered. It also enabled the research group to triangulate stakeholder evidence derived from multiple sources.

We read a confidentiality agreement at the beginning of each workshop. Two-hour discussions followed a semi-structured guide and were recorded and transcribed with participants' permission (see SM for a copy of the guide). Afterwards they were provided with a report summarising their workshop discussions, which they approved prior to analysis.

2.2.1.5. Review seminars. We held four review seminars across Scotland to jointly discuss preliminary findings from our literature review, questionnaire and focus groups to initiate constructive discussion of possible solutions to the conflict. Seminars were open to anyone interested in the project, but we especially encouraged participation among questionnaire respondents and those representing views not widely shared by other attendees. Participants with different interests were seated next to each other to encourage cross-sector relationship-building. In the final seminar, stakeholders mutually prioritised and discussed 29 future collaborative actions building on results from a short online survey we developed of ideas emerging from the project.

2.3. Analysis

Transcriptions of qualitative data from the questionnaire, focus groups and seminars were imported into NVivo software (V10, QSR International Pty). We used an open coding approach to manually sort information according to key themes important to the study that we identified from literatures spanning transdisciplinary research, knowledge co-design and co-production, co-management, and participation. The five most relevant themes were: integrating different knowledge types, transparency, learning and exchange, trust, and conflict transformation (see SM Table 1).

3. Results

3.1. Integrating different knowledge systems

Nine in ten (89 %) questionnaire respondents affiliated with one or more of ~70 game management, ornithological, conservation or scientific organisations/professions. We used an iterative approach to classify respondents into ‘local’ (40 % of respondents) and ‘scientific’ (27 %) knowledge groups, according to the forms of information they identified in the questionnaire as most important when learning about predation of ground-nesting wild birds. Practitioners strongly favoured local knowledge (e.g. personal field observations) whereas representatives of conservation organisations favoured scientific knowledge (e.g. scientific peer-reviewed studies). However, many respondents claimed to rely on both in some circumstances. We similarly classified focus groups into six local and three scientific knowledge groups.

Focus group discussions revealed strong agreement between scientific and local knowledge about changes in focal species abundance and drivers of population change. This led to some participants perceiving themselves more confidently as crucial actors in the management of ground-nesting bird populations and expressing a desire to collaborate with those holding other kinds of knowledge where previously they had not wanted to: *“...our local knowledge tied in with competent scientific methodology and thinking, what a wonderful way forward...”* (local knowledge group participant).

Reflecting on this agreement between the knowledge sources, seminar participants considered that both local and scientific knowledge were needed in the study context and that the project had been useful in bringing together a broad range of stakeholders: *“It’s leading to better communication between stakeholders and they are beginning to understand each other’s approaches”* (seminar participant). The combination of scientific literature reviewing, collection of local knowledge, and the opportunity to explore areas of agreement and disagreement were seen as useful and effective: *“The results have been eye-opening”* (seminar participant).

Some participants suggested that future knowledge integration could build on the strengths of the different types of knowledge and address their weaknesses, but the need for early involvement of stakeholders was noted: *“...there’s a key issue about design so that people providing local knowledge feel they’ve been asked the right questions and equally that the science is able to take on local knowledge and compare it in a way people can see is a valid and fair comparison and neither side is biased”* (seminar participant).

3.2. Transparency

A key aspect linked to transparency was the nature of local knowledge. Based on the results of the questionnaire, local knowledge tended to be transmitted verbally within peer groups rather than written down or published and was therefore not accessible to the broader community in the same way that scientific knowledge was available through publications. Participating in a focus group was an opportunity to have individual and organisational values and knowledge formally recorded and made publicly available via the report.

Several group participants appreciated the opportunity to share their knowledge and perspectives about the subject matter and the conflict situation in an in-depth, non-confrontational way: *“No one from our industry, no one’s willing to talk about the taboo subject of raptors and the issues that we have with them. So, for me this is a step forward; it’s brilliant”* (local knowledge group participant). Another local knowledge group participant remarked that: *“We’re really pleased that this [UP Project] is being done and hopefully something will come out of this that will reflect our views on this... the predatory bird report [Park et al., 2005]... went in the bottom of a cabinet and got locked away”*.

Others felt that reporting on knowledge held by diverse

stakeholders involved in the project could help debunk what some described as ‘rural myths’ disseminated by people they considered among the more extreme elements of the community of interest: *“We’ve got the gamekeeper, the hunting side; we’ve got outsider conservationists and so on. In the middle is a vast, vast majority of the general public who have no real knowledge or education in conflicts that occur. I think one of the big outputs should be education of the public because at the end of the day if there’s going to be any legislation... maybe they’re going to vote on it”* (scientific knowledge group participant).

3.3. Learning and exchange

Research materials and methodologies were developed based on mutual learning and prioritisation of research objectives derived from planning, steering and research group discussions about the project context and stakeholder requirements. This ongoing ‘peer-review’ process allowed the research group to continually question their own and others’ assumptions regarding the research scope and objectives to ensure that salient concerns were identified and addressed while the scope of the project remained tenable. Distributing accountability for decision-making across diverse interest groups meant that many of those who joined the project holding opposing views recognised a mutual desire to jointly develop sustainable outcomes. Social learning among stakeholders arising from their contribution to the research program and reflection on the integrated scientific and local knowledge results enabled participants in the fourth seminar to mutually prioritise future collaborative projects.

Nevertheless, some members of the stakeholder community and some research participants raised concerns about the design of the project. Some participants relying on scientific knowledge suggested that the project focussed too much on the negative impacts of predation, rather than broader ecosystem / landscape processes. This led to some confusion about the project’s overall purpose: *“...we don’t really understand what the output of the project is”* (scientific knowledge group participant). We attempted to address this by emphasising that future actions should integrate both practical and theoretical elements to encourage those holding scientific and/or local knowledge to collaborate.

3.4. Trust

Some local knowledge holders distrusted reporting of scientific evidence, mostly due to perceived biases in the funding, peer-review, framing and spatial-temporal scale of studies: *“...when you’ve got somebody that’s throwing money at the science and you know that what they’ve put out is absolutely rubbish by your observations you cannot trust that science”* (local knowledge group participant).

A few scientific knowledge holders criticised what they saw as a historical reliance on circumstantial evidence among practitioners: *“I respect and value people’s field experience greatly, but it is very easy to reach conclusions and be very firmly of a fixed point of view on things based on personal experience, but still be wrong”* (scientific knowledge questionnaire respondent). This related to a distrust of non-scientific sources, such as news media and non-peer-reviewed literature written by non-government organisations, which they said was sometimes politically motivated or influenced by members’ interests.

Seminar participant feedback was generally positive due to the project’s approach and methods. These were designed to encourage collaboration on developing creative solutions to shared problems, and were seen as having the potential to increase trust among individuals. Nevertheless, seminar participants noted the need to continue building trust through increased links: *“The key will be building trust and respect: this will require breaking down barriers, for example by spending more time together to see how each group works”* (seminar participant).

Some seminar participants said that integration of different forms of knowledge could be achieved by increasing opportunities for networking and creating an appropriate group to integrate knowledge in

areas where there was disagreement and bring agreed findings to policy makers. This suggests that trust at a societal and intermediate level was still somewhat unrealised and was not adequately supported via the Forum.

3.5. Conflict transformation

A critical realisation emerged through repeated dialogue that conservation-management of the ground-nesting birds was a fundamental desire among all stakeholder groups, and that the underlying conflict lay in whether stakeholders preferred applying habitat or predator management to achieve sustainable bird populations. Reaching a shared definition of how stakeholders perceived ‘healthy populations’ of predator and prey species was also critical. Definitions were associated with achieving an overall balance and/or sustainability of both predator and prey populations allowing for fluctuating populations of both but stability over the longer term. However, ideas of balance and sustainability also varied between the groups, reflecting their management objectives.

The focus groups provided a forum for participants to voice their opinions about various aspects of the contentious context within which they were working. Many used this as an opportunity to criticise government agencies or organisations with whom they were in conflict or to discuss perceived failures in the current regulatory system for managing predators and prey. However, many local and scientific participants also suggested potential collaborative solutions to common problems and demonstrated a willingness to cooperate, which may not otherwise have been apparent to external observers: “*I’d love to be able to work with RSPB, as we do with the British Trust for Ornithology and other scientists...*” (local knowledge focus group).

Many seminar attendees noted benefits in the project of bringing the various interest groups together and providing them with opportunities for discussion. There was a shared aspiration amongst stakeholders to find ways of maintaining this positive dialogue and create opportunities to work together through future collaborative actions. On completion of the UP project, Moorland Forum stakeholders developed the ‘Working for Waders’ program which builds upon these shared goals (see <https://www.moorlandforum.org.uk/working-for-waders>).

4. Discussion

To contextualise our results, we compared and integrated five complementary theoretical approaches: knowledge co-production, community science, knowledge integration, knowledge implementation, and conflict transformation. These concepts integrated well and overlapped frequently suggesting the novel set of emerging considerations provides consistent, valuable guidance for achieving knowledge integration goals when mapping conservation conflicts (see Fig. 4 for a summary and SM Table 2 for full details). Below, we discuss lessons learned from our experience in the UP project, according to these considerations.

4.1. Lessons learned: what worked well

Two recognised end points of research integration are: 1) developing a more comprehensive understanding of the problem, plus possible ways to address it, by integrating disciplinary and stakeholder perspectives; and 2) supporting implementation of that understanding into evidence informed policy and practice (Bammer et al., 2020). Success is indicated by the extent to which participants use process outputs (Raymond et al., 2010).

The UP project achieved both end points. Our iterative and transparent approach to working on an equal footing with the steering group members is similar to that of adaptive management and governance strategies whereby we prioritised communication, perspective sharing, social learning, negotiation, and the development of adaptive

collaborative strategies for moving forward (Berkes, 2017). For example, given the high level of uncertainty about the subject matter, our integrated methods provided a flexible space where we could respond to changing conditions and allowed for learning to be captured and incorporated into subsequent rounds of deliberation and analysis (Raymond et al., 2010; Bracken et al., 2015; Berkes, 2017).

Through social learning, participants significantly advanced their understanding of the local and scientific knowledge held among them and where/why perspectives aligned and conflicted. This enabled participants to view aspects of each other’s knowledge as more credible and legitimate than previously and to agree on the best available science which they ultimately applied to identify joint actions that they could and have put into place (through Working with Waders; Section 3.5) (Cravens and Ardoin, 2016). Indeed, social learning was crucial to our knowledge integration process as it facilitated mutual acceptance of diverse positions and greater openness to collaboration (Reed et al., 2010). Several conditions helped us achieve these end points.

First, the project was *context-based*, being community-driven and community controlled through its connection to place and community values (Charles et al., 2020; Norström et al., 2020). We benefited from links to governance through a clear mandate from government via the Moorland Forum, which acted as a bridging organisation spanning the boundary between science and management (Berkes, 2009; Cook et al., 2013; Norström et al., 2020). The Forum provided access to capacity and resources. Representatives from eight Forum member organisations formed a very active steering group, representing different knowledge systems and stakeholder needs, who advised the research group from framing the research question to joint review of results.

The research group held context-relevant expertise in research integration and implementation in conservation conflict situations (Bammer et al., 2020). They worked on an equal footing with the steering group to identify clear goals, bring different stakeholders together and negotiate indicators of success (Ludwig et al., 2001; Redpath et al., 2013). They successfully identified appropriate methods and implementation pathways to recognise and integrate important insights into the project findings (Bammer et al., 2020). The group was mostly successful in navigating the conflict and handling stakeholder expectations throughout the project by managing contested problem definitions (defining healthy populations of predators and prey); delimiting the problem and managing critical, un-resolvable unknowns (focusing on knowledge about ground-nesting birds); and accommodating the partial and temporary nature of solutions (helping stakeholders identify future collaborative actions) (Bammer et al., 2020).

By working collectively and building on the process and outcomes of this project, participants with local and scientific knowledge developed integrative and strategic knowledge, thereby strengthening their tactical positions (Aslin and Brown, 2004). Those holding long-term adversarial positions learned to negotiate over common priorities and improved the technical quality of decisions through the mutual prioritisation of future collaborative actions. Thus, participants achieved several known benefits of community science approaches: developing comprehensive understanding and implementing joint next steps to address a problem; creating learning opportunities to cope with complexity and uncertainty; and building community resilience (Charles et al., 2020). For example, in identifying future collaborative actions, participants became more flexible and forward-looking as a way of adapting and beginning to transform the conflict (Berkes, 2017). By embarking on the collaborative actions in Working for Waders, participants opened themselves to the possibility of learning and adjusting their responses to the impacts of external drivers and internal change within their stakeholder communities (Berkes, 2017).

Second, we took a *pluralistic* representation and engagement approach (Reed et al., 2014; Norström et al., 2020). Mainly thanks to the steering group and community-led nature of the project, numerous diverse stakeholders engaged with the research. Research group members’ methodological expertise facilitated implementation of multiple

Source	Principles and conditions
Evaluating knowledge co-production processes (Norström et al., 2020)	Context-based; pluralistic; goal-oriented; interactive
Principles /conditions of community science (Charles et al., 2020)	Community-driven and community-controlled; connection to place and collective values; links to governance; leadership; empowerment, agency and collective action; availability of capacity and resources
Knowledge integration and implementation expertise (Bammer et al., 2020)	Manage different start points; identify relevant discipline-based researchers and stakeholders; integrate useful perspectives; assess and combine suggestions for action, determine strengths and risks, identify implementation options and pathways; delimit the problem; manage contested problem definitions; manage critical, un-resolvable unknowns; manage real world constraints on ameliorating the problem; appreciate and accommodate the partial and temporary nature of solutions
Principles of conflict transformation (Redpath et al., 2013)	Clear goals; identifying what different stakeholders perceive as success; engagement: bringing parties face to face to discuss and negotiate acceptable solutions; engagement is important for prioritising conflict mitigation techniques and for design and implementation of the conflict transformation process; transparency in the process so that stakeholders can participate in discussions on an equal footing and make informed decisions; engagement contributes to parties reaching good-quality agreements and improved relationships; the outcome is acceptable to both sides and neither party is asserting its interests to the detriment of others
Key knowledge integration themes (Supplementary Material Table 1)	Transparency; integrating different knowledge systems; learning and exchange; trust; conflict transformation

Fig. 4. Principles and conditions relating to the development of effective community-led, knowledge co-production and integration processes in conservation conflict situations.

interacting engagement strategies to create the conflict mapping process around integrated stakeholder knowledge (Redpath et al., 2013). This helped address known pitfalls regarding access to knowledge and communication of personal rather than institutional perspectives (Sillitoe, 2014; Sillitoe and Filer, 2014).

We treated all stakeholder views in a democratic and accountable manner and attempted to address uneven power relations by creating scope to highlight differences and enable contestation of interests, views and knowledge claims (St Clair, 2016; Turnhout et al., 2020). The project demonstrated how to bring diverse interest groups together to discuss divisive topics in an intelligent, considered and constructive fashion, paving the way for learning and trust to develop (Cornell et al., 2013). However, to keep the process manageable and encourage ownership of the research problem, sometimes it was important to limit the number of participants, for example the focus groups included stakeholders most directly affected by the conflict (Lamers et al., 2010).

Participation in the planning and steering groups allowed some individuals to work more closely together than before and to get to know each other better, and such relationship building was crucial to the success of the process (O'Connor et al., 2019). Maintaining and building trust both during this project and after was essential if stakeholders were to have confidence in the research process and conflicts around predation were to transform successfully (Beierle and Konisky, 2001; Pearce, 2008; Stepanova, 2014; Young et al., 2016). Of benefit was that some stakeholders trusted research group members who were already familiar with the project context and similar kinds of conflict situations or perceived as impartial by those concerned.

Third, our research was strongly *goal-oriented* (Raymond et al., 2010; Redpath et al., 2013; Reed et al., 2014; Norström et al., 2020). We mutually identified a shared problem which was concern about declining ground-nesting bird populations. We were mostly successful in defining for the first time the extent of the community of interest; the breadth of opinions held about the topic; and the depth of participants' knowledge and experience of predation. For some stakeholders, discovering where local and scientific knowledge overlapped or was complimentary regarding the shared problem engendered empowerment, agency and collective action as it helped lessen disputes and

move the conversation on to other priorities, such as identifying future collaborative actions and developing the Working for Waders program (Charles et al., 2020).

Finally, our approach was *interactive* (Reed et al., 2014; Norström et al., 2020). The formal nature of stakeholder partnerships among those governing the project meant that regular, transparent communication was essential. Throughout, members of the research and steering groups maintained continuous, open and iterative dialogue with stakeholders and each other about project aims, objectives, capacities, and reasons and motivations for actions. This approach was crucial to avoid hidden agendas, guarantee equal weight in decision-making, ensure decisions reflected public values and effectively work towards social change (Beierle and Konisky, 2001; Jepson and Canney, 2003; Aslin and Brown, 2004; Durham Community Research Team, 2011; McCarthy and Fluck, 2017). Over time, participants became more transparent with us, and with one another, regarding their value positions and the knowledge they held and learned more about their own and each other's values, knowledge and motivations as well as about the participatory research process.

Members of the community of interest could share and exchange relevant knowledge amongst themselves and with the general public and reflect on their social realities through the mixed methods approach, which provided a means for social learning (Reed et al., 2010). Where there were commonalities or differences in understanding, participants could potentially increase their appreciation for other perspectives and ways of knowing. Thus, they were able to develop some trust based on confidence, legitimacy, shared values and appropriate behaviour, as noted elsewhere (e.g. Beierle and Konisky, 2001; Aslin and Brown, 2004; Young et al., 2016).

4.2. Lessons learned: what didn't work

Despite carefully framing the project and approaching diverse knowledge holders, we could not capture all relevant stakeholders. Certain important organisations did not fully participate due to concerns about the project design. Important farming communities did not participate despite ongoing attempts to engage the National Farmers

Union Scotland (a Forum member) in early discussions. This is likely because the issues framed by UP initiatives were not priorities in the same way they were for gamekeepers and conservationists whose values were in conflict and thus most salient among stakeholders and the public. The farmers' self-censorship alludes to the asymmetrical power relations evident among stakeholder organisations regarding legitimacy of their knowledge and availability of time and resources to participate, and potentially lessened the usefulness of the project outcomes for them (Norström et al., 2020; Turnhout et al., 2020).

With those stakeholders who were engaged, significant effort was required to ensure that all voices were acknowledged, respected and listened to throughout the process (Aslin and Brown, 2004). Community leadership was crucial to linking the project's goals to stakeholders' existing priorities and activities (Charles et al., 2020; Norström et al., 2020) and the importance of a trusted chair facilitating meetings could not be over-emphasised. Indeed, the issue of trust was crucial throughout the process and we risked a loss of trust if one or more stakeholders or knowledge groups felt we did not adequately meet their expectations (Beierle and Konisky, 2001; Young et al., 2016). To illustrate, we did not manage to reach consensus on the appropriate weighting of key values central to the study issues held by scientific and local knowledge holders (Douglas, 2009), resulting in criticism of our questionnaire and report by some stakeholders. This highlights the different conceptions of success existing among our project participants which we could not adequately reconcile (Raymond et al., 2010; Norström et al., 2020).

4.3. Towards a practical framework for realising knowledge integration goals in conflict situations

Based on our findings, we present a novel, practical framework aimed at individuals and groups working in co-production processes for realising knowledge integration goals in situations where deeply held values, knowledge systems and worldviews could benefit from

knowledge integration processes (Fig. 5). The process builds on the five synthesised theoretical approaches described above and successful implementation requires embracing throughout the considerations summarised in Fig. 4.

Constant monitoring and improvement through a transparent, iterative and goal-oriented 'learning by doing' approach (Pearce, 2008) is advisable so that different perspectives can continuously inform technical decisions about the design, oversight and outputs of the integration process. Such monitoring should allow for pro-active recognition of whether parties want to pull out of the collaboration, in which case genuine effort must be made to understand the reasons why they may not be engaging in the process, and to address those reasons. Over time, the above approach should engender social learning and build trust among stakeholder groups, thereby potentially helping to transform the conflict (Ramsbotham et al., 2014; Young et al., 2016).

Finally, we concur with Raymond et al. (2010) that the process should focus on a clear problem and outcome of relevance and importance to all stakeholders. In this study, the identification of future collaborative actions was a major outcome for many stakeholders. However, this needs to be followed up by strong voices to government to encourage action on areas where those relying primarily on scientific or local knowledge agree; and co-operating in further research or other activities to further integrate knowledge and/or management outcomes.

5. Conclusions

Conservation conflicts between humans pose serious and persistent challenges to international wildlife conservation due to underlying social factors. We argue that in situations where there is disagreement over the evidence base, from the outset efforts can and should be made to integrate not just scientific and local knowledge, but also strategic knowledge held by participants and researchers, and all relevant parties should be involved in a genuine manner throughout the process.

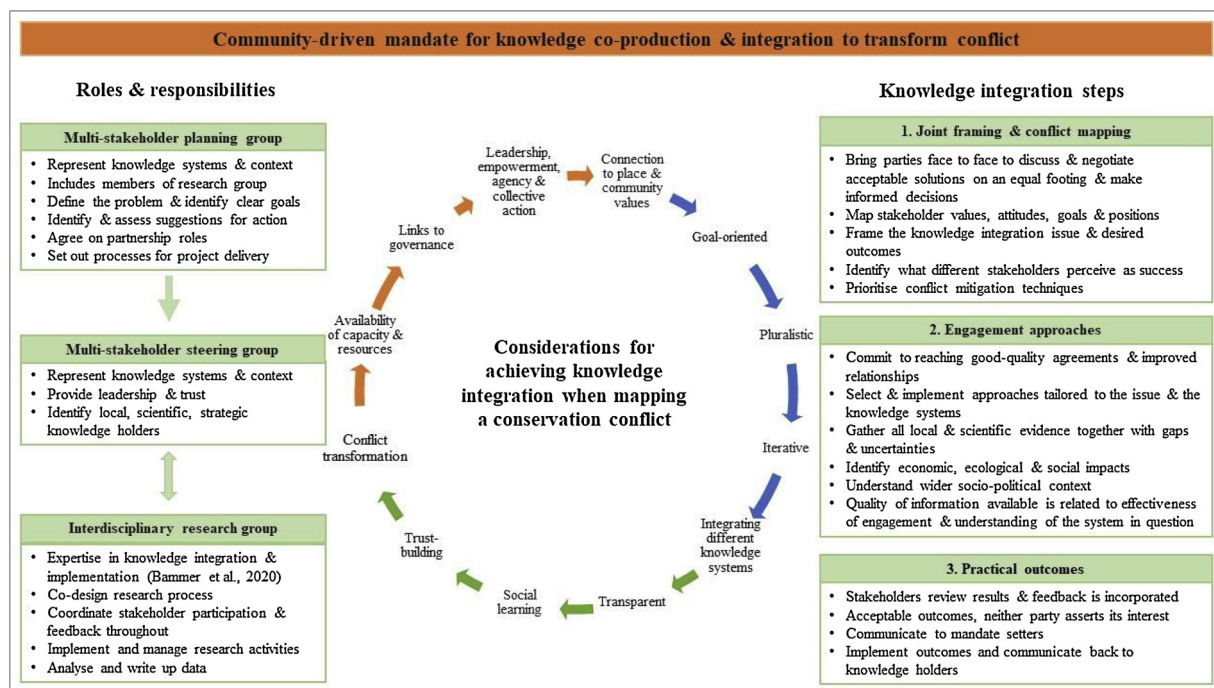


Fig. 5. A framework for the practical integration of diverse knowledge systems when mapping conservation conflicts. The process starts with a community-driven mandate for conflict transformation. Boxes (left) represent roles, responsibilities and expertise of groups regarding process design and implementation. Boxes (right) represent important steps in the process. The cycle (centre) represents the dynamic and inter-linked nature of considerations for conflict transformation based on principles of community science (orange arrows, Charles et al., 2020), knowledge co-production (blue arrows, Norström et al., 2020) and knowledge integration processes (green arrows, Table 1, this paper), and are a fundamental ethic of the entire process.

Studying and mapping conservation conflicts using our framework can provide a means to better understand community motivations and the conditions conducive to collective action. However, this requires a shift in thinking about how to approach environmental governance. Advice on implementing such approaches is lacking and we believe no single participatory approach offers the tools required to foster conflict transformation.

Our research highlighted the importance of community science because the motivation to transform conflict must come from affected community members if they are to commit to the lengthy and challenging process of building trust and collaborating on mutual goals. Knowledge co-production is important because the diverse values and priorities of community members must be recognised through a context-based, pluralistic, goal-oriented, interactive and social learning approach. Expertise in knowledge integration and implementation is vital for applying appropriate tools and techniques to successfully implement a process within real world constraints. Finally, researchers should have experience in conducting conflict transformation processes to manage stakeholder expectations while helping them achieve good quality agreements and improved relationships thereby moving the conflict in a positive direction.

To conclude, our research contributes to an important academic and policy gap as it is the first attempt to synthesise several theoretical approaches to conduct and evaluate a community-led, knowledge co-production and integration process in a conservation conflict situation. By providing advice on important considerations to address known challenges, our innovative, practical framework can guide researchers, decision-makers and different kinds of knowledge holders to prioritise their goals and successfully develop an ethical and robust knowledge integration process to transform conservation and other kinds of conflict situations. Our framework and considerations can also inform processes such as IPBES which currently struggle to retain the authenticity of local knowledge when merged with scientific knowledge.

Compliance with ethical standards

This project did not raise any non-routine ethical considerations. The research was not carried out with vulnerable populations and did not involve participants being subjected to questions which carry a risk of being harmful to their well-being. Participation in questionnaires, focus groups and seminars was confidential and results were anonymised to protect individual privacy. Participant consent was sought to record focus groups prior to them taking place. No conflicts of interest regarding author contributions occurred.

CRedit authorship contribution statement

Gillian Barbara Ainsworth: Methodology, Writing - original draft, Data curation, Visualization, Investigation. **Stephen Mark Redpath:** Conceptualization, Supervision, Writing - review & editing, Funding acquisition, Investigation. **Mark Wilson:** Conceptualization, Writing - review & editing, Funding acquisition, Investigation. **Chris Wernham:** Conceptualization, Writing - review & editing, Funding acquisition, Investigation. **Juliette Claire Young:** Conceptualization, Methodology, Supervision, Investigation, Funding acquisition, Writing - original draft.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.envsci.2020.02.017>.

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